

Ideal gas model

Learning outcomes

By the end of this section you should be able to recognise the key assumptions in the ideal gas model.



Figure 1. A bottle filled with gas can be compressed.

Credit: Proximo, Getty Images

(<https://www.gettyimages.com/detail/photo/crushed-plastic-bottle-royalty-free-image/1302610076>)

Have you ever wondered why air can be compressed, but not ice? When you think of the main difference between these two substances, you may realise that air is a gas, and ice is a solid. Air consists of a mixture of gases, mainly nitrogen gas, N_2 , and oxygen gas, O_2 . How is the behaviour of particles different in gases when compared to solids and how can it help explain the fact that gases can be compressed but not solids?

Practical skills

Inquiry 1: Exploring and designing — Controlling variables

Inquiry 3: Concluding and evaluating — Evaluating

Give the following a try:

- Gather three plastic water bottles.
- Keep the first one empty and close it.
- Fill the second one with water and close it.
- Fill the third one with water, close it and place it in the freezer overnight.

Try squeezing the three bottles. Which can be compressed the most? The least? Why? How does this help explain the arrangement of particles in solids, liquids, and gases?

In section S1.1.2a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>), we looked at the properties of the different states of matter (solids, liquids and gases). Here, we will look in more detail at the properties of gases, specifically the kinetic molecular theory of gases, which outlines the assumptions of the ideal gas model:

- The particles in a gas are in constant, random, straight-line motion.
- There are negligible forces of attraction (intermolecular forces) between the particles.
- Collisions between particles or with the walls of the container are perfectly elastic (no energy is lost).
- The distance between the particles is much greater than the size of the particles, therefore, gas particles have negligible volume.
- The average kinetic energy of the particles in a gas is directly proportional to the absolute temperature (in kelvin). This means that if an ideal gas at a temperature of 100 K is heated and its temperature doubled to 200 K, then it is assumed that the kinetic energy of the gas particles will also double. If the temperature is halved, then the kinetic energy is halved, etc.

As a gas has no defined shape or volume, it must be contained within a container. The quantity of gas inside the container will be influenced by several variables – volume, temperature and pressure. Volume is often measured in millilitres, expressed as mL or cm^3 , or litres, expressed as L or dm^3 . Temperature is measured by the absolute temperature in units of kelvin. Pressure is determined by how frequently the particles hit the walls of the container and is expressed in Pa or kPa (**Figure 2**).

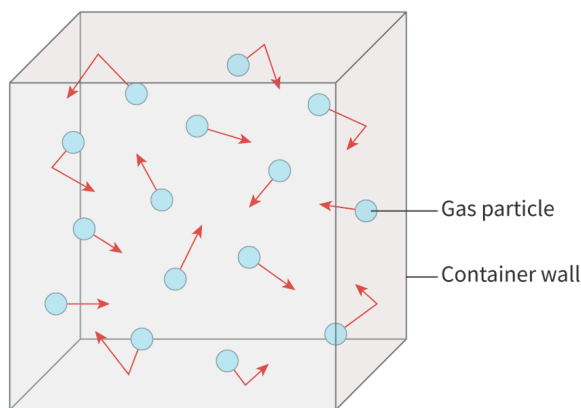


Figure 2. The pressure of the air in a container depends on the frequency with which the particles hit the wall — it is independent of the nature of the particles.

© More information for figure 2

The model proposed by the kinetic molecular theory suggests that, in an ideal gas, the particles are very far apart and moving randomly. In a sample of a gas most of what is there is actually empty space. This predominance of empty space in gas results in the assumption that the size of the gas particle and the interaction of the particles are negligible. The volume of these gases under the same conditions will be the same. Therefore, you can have samples of gas with the same volume, but a different number of gas particles inside because of the empty space. Imagine a school with three classrooms identical in size to the one you are sitting in right

now. There might be 20 students inside one classroom, 10 in another and just 3 in the last classroom. All of the classrooms have the same volume, but can accommodate different numbers of students – there is just less space between the students. In which classroom will there be more collisions between the walls and other students? What would happen if you introduced even more students to each classroom?

The ideal gas model also proposes that the attractions between molecules can be assumed to be non-existent, and any collisions between them can be assumed to be elastic. In the analogy above, all of these students can move around freely and if they hit the wall or another student, they just bounce off, without losing any energy or getting tired.

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

The air  (<https://climate.nasa.gov/news/2491/10-interesting-things-about-air/#:~:text=Air%20is%20mostly%20gas&text=It's%20a%20mixture%20of%20different,dioxide%2C%20neon%2C%20>) we breathe is a mixture of gases, mainly nitrogen and oxygen amongst others. Air pollution is a major environmental problem and **WHO data**  (https://www.who.int/health-topics/air-pollution#tab=tab_1) suggests that up to 99% of the world's population is exposed to some form of air pollution.

How can you as an IB student help reduce the amount of air pollution produced by your school or in your local community? One possible way is to conduct an energy audit at your school and make a list of suggestions of the ways in which the amount of air pollution produced by your school can be reduced. You can download the [guidance sheet](https://19january2017snapshot.epa.gov/sites/production/files/2016-07/documents/wheel_studentact3.pdf)  (https://19january2017snapshot.epa.gov/sites/production/files/2016-07/documents/wheel_studentact3.pdf) from the Environmental Protection Agency (EPA) for ideas on how to conduct an energy audit of your school.